

Computational Algebraic geometry, Applications to Physics, and Macaulay2

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Abstract. Theoretical physics and string theory provide useful hard problems to solve involving polyhedra geometry, computational algebra and computational algebraic geometry. We describe some of these problems, some solutions, and several challenges and open problems for the computational algebra and algebraic geometry community. The work described here is based on joint work with Liam McAllister (Cornell; string theory and cosmology) and several other physicists.